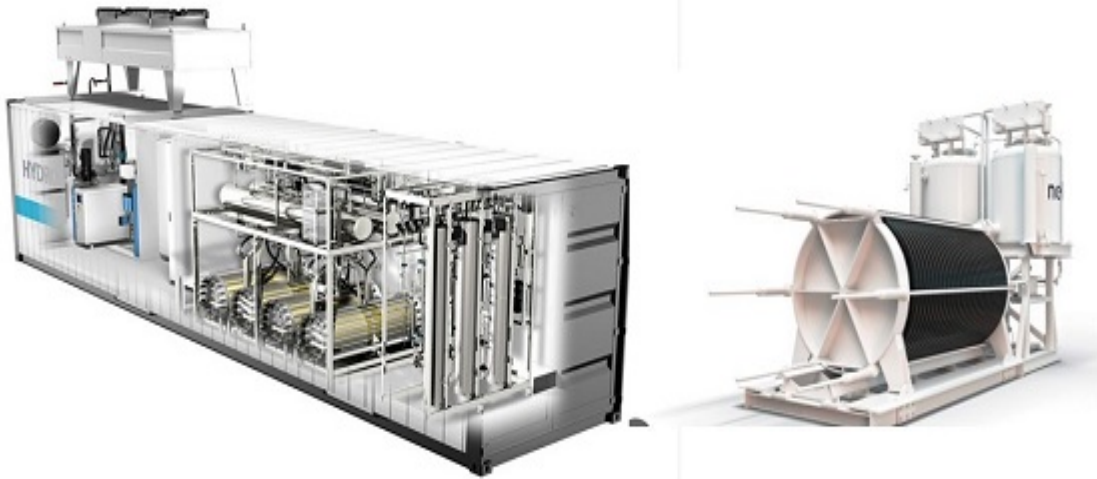


Qualification Pack



Electrolyzer Manufacturing Plant Technician

QP Code: SGJ/Q4306

Version: 1.0

NSQF Level: 4

Skill Council for Green Jobs || 3rd Floor, CBIP Building, Malcha Marg, Chanakyapuri
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Qualification Pack

Contents

SGJ/Q4306: Electrolyzer Manufacturing Plant Technician	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
SGJ/N4318: Introduction about the Electrolyzer and its types	5
SGJ/N4319: Identify Electrolyzer Components and Materials	10
SGJ/N4320: Perform Manufacturing Process of Electrolyzer	14
SGJ/N4321: Identify Tools and Tackles used in Electrolyzer Manufacturing Facility	19
SGJ/N4322: Perform Health and safety measures in Electrolyzer manufacturing facility.	24
DGT/VSQ/N0102: Employability Skills (60 Hours)	29
Assessment Guidelines and Weightage	36
<i>Assessment Guidelines</i>	36
<i>Assessment Weightage</i>	37
Acronyms	38
Glossary	39

Qualification Pack

SGJ/Q4306: Electrolyzer Manufacturing Plant Technician

Brief Job Description

Electrolyzer Manufacturing Plant technician is responsible for supporting the safe, efficient, and high-quality production of electrolyzer system. Their work is essential to provide technical support to maintain the manufacturing process of electrolyzer and meeting production goals while adhering to safety and environmental standards. He/She plays a vital role in producing and maintaining electrolyzer along with associated subsystems and is accordingly responsible for various electrical components and systems-related tasks within the Electrolyzer manufacturing facility.

Personal Attributes

This unit tells about job role of technician in Electrolyzer Manufacturing Plant.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N4318: Introduction about the Electrolyzer and its types](#)
2. [SGJ/N4319: Identify Electrolyzer Components and Materials](#)
3. [SGJ/N4320: Perform Manufacturing Process of Electrolyzer](#)
4. [SGJ/N4321: Identify Tools and Tackles used in Electrolyzer Manufacturing Facility](#)
5. [SGJ/N4322: Perform Health and safety measures in Electrolyzer manufacturing facility.](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Green Jobs
Sub-Sector	Green Hydrogen
Occupation	Green Hydrogen
Country	India
NSQF Level	4
Credits	14

Qualification Pack

Aligned to NCO/ISCO/ISIC Code	NCO-2015/ 8131.2100 Ammonia Operator/Ammonia Plant Operator
Minimum Educational Qualification & Experience	12th grade Pass (Or Equivalent) with NA of experience OR Pursuing 3rd year of 3-year diploma after 10th (Electrical/ Mechanical/Chemical) with NA of experience OR Previous relevant Qualification of NSQF Level (3.5) OR Previous relevant Qualification of NSQF Level (3)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	31/01/2027
NSQC Approval Date	31/01/2024
Version	1.0
Reference code on NQR	QG-04-ES-02027-2024-V1-SCGJ
NQR Version	1

Remarks:

Total 420 Notional Hours: (Theory: 220 hours with Employability Skills+Practical:140 hours+ 60 hours of On-the-Job Training (OJT)

Qualification Pack

SGJ/N4318: Introduction about the Electrolyzer and its types

Description

This unit tells about Introduction about the Electrolyzer and its types

Scope

The scope covers the following :

- Describe Electrolysis Process.
- Explain about Electrolyzer Technology.
- Discuss about applications of Electrolyzer.
- Introduce different types of Electrolyzer

Elements and Performance Criteria

Basics of electrolysis, electrolysis process and Electrolyzer technology

To be competent, the user/individual on the job must be able to:

- PC1.** describe and demonstrate electrolysis process on a small model
- PC2.** explain and demonstrate working of electrolytic cell through picture or videos.
- PC3.** explain and demonstrate laws related to electrolysis.
- PC4.** discuss in brief various terms associated with Electrolysis Process.
- PC5.** discuss and illustrate about Electrolyzer technology through pictures and videos
- PC6.** discuss key aspects and challenges related to electrolyzer manufacturing in India.
- PC7.** discuss about the factors affecting electrolysis.
- PC8.** discuss and demonstrate the roles of a technician at an Electrolyzer Manufacturing facility

Explain different types of Electrolyzers and technical specification.

To be competent, the user/individual on the job must be able to:

- PC9.** discuss and illustrate the classification of Electrolyzer through pictures or videos
- PC10.** discuss and illustrate about alkaline electrolyzer technology along with technical specifications of Alkaline electrolyzer.
- PC11.** discuss and illustrate about Proton exchange membrane (PEM) electrolyzer with technical specifications.
- PC12.** discuss and illustrate the technical specifications of Solid oxide electrolysis cell (SOEC) electrolyzer.
- PC13.** discuss Anion exchange membrane (AEM) electrolyzer with technical specifications.
- PC14.** discuss Pros and Cons of different Electrolyzer types.
- PC15.** discuss about advancement in Electrolyzer Technology and Future of Electrolyzer.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** legislative, organizational, site requirements and related procedures
- KU2.** usage and handling procedure of system components
- KU3.** different mechanical and electrical equipment, their functioning and precautions
- KU4.** importance of wearing protective clothing and other safety gear
- KU5.** tools involved in installation and O&M and manufacturing activities
- KU6.** reporting procedures
- KU7.** cleaning procedures for various equipment

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** undertake complete documentation
- GS2.** correctly read and interpret various color codes, as per standard electrical, mechanical and civil nomenclature
- GS3.** follow organizational rule-based decision making process
- GS4.** approach relevant authority when required
- GS5.** work constructively and collaboratively with others
- GS6.** express statements with required clarity
- GS7.** make timely decisions for efficient utilization of resources

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Basics of electrolysis, electrolysis process and Electrolyzer technology</i>	20	30	-	-
PC1. describe and demonstrate electrolysis process on a small model	2	6	-	-
PC2. explain and demonstrate working of electrolytic cell through picture or videos.	2	6	-	-
PC3. explain and demonstrate laws related to electrolysis.	2	6	-	-
PC4. discuss in brief various terms associated with Electrolysis Process.	4	-	-	-
PC5. discuss and illustrate about Electrolyzer technology through pictures and videos	2	6	-	-
PC6. discuss key aspects and challenges related to electrolyzer manufacturing in India.	2	-	-	-
PC7. discuss about the factors affecting electrolysis.	3	-	-	-
PC8. discuss and demonstrate the roles of a technician at an Electrolyzer Manufacturing facility	3	6	-	-
<i>Explain different types of Electrolyzers and technical specification.</i>	20	30	-	-
PC9. discuss and illustrate the classification of Electrolyzer through pictures or videos	2	6	-	-
PC10. discuss and illustrate about alkaline electrolyzer technology along with technical specifications of Alkaline electrolyzer.	3	7	-	-
PC11. discuss and illustrate about Proton exchange membrane (PEM) electrolyzer with technical specifications.	3	7	-	-
PC12. discuss and illustrate the technical specifications of Solid oxide electrolysis cell (SOEC) electrolyzer.	3	10	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. discuss Anion exchange membrane (AEM) electrolyzer with technical specifications.	3	-	-	-
PC14. discuss Pros and Cons of different Electrolyzer types.	3	-	-	-
PC15. discuss about advancement in Electrolyzer Technology and Future of Electrolyzer.	3	-	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4318
NOS Name	Introduction about the Electrolyzer and its types
Sector	Green Jobs
Sub-Sector	Green Hydrogen
Occupation	Hydrogen Plant Technician
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

SGJ/N4319: Identify Electrolyzer Components and Materials

Description

This unit tells about identification of Electrolyzer Components and Materials

Scope

The scope covers the following :

- Raw material Required for Electrolyzer Manufacturing.
- Identify Bill of material for materials and components of Electrolyzer manufacturing set up.

Elements and Performance Criteria

Materials and components required for Manufacturing Electrolyzer

To be competent, the user/individual on the job must be able to:

- PC1.** explain and illustrate the basic materials and components required for manufacturing Electrolyzer and different types of Materials required for Electrolyzer manufacturing and their key specification through Pictures, videos, product data sheet etc.
- PC2.** explain the major components required for Alkaline type Electrolyzer. and discuss about each component.
- PC3.** explain the major components required for PEM type Electrolyzer and discuss about each component
- PC4.** explain the major components required for (SOEC) type Electrolyzer and discuss about each component.
- PC5.** prepare and Identify Bill of material for materials and components of Electrolyzer manufacturing set up.
- PC6.** explain the major components required for (AEM) type Electrolyzer and discuss about each component.
- PC7.** outline differences in PEM, AE, Electrolyzer Materials and components.
- PC8.** show how to identify Bill of material for materials and components of Electrolyzer manufacturing set up.
- PC9.** showcase the successful case studies of Electrolyzer technology for Green Hydrogen generation in India and overseas.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislative, organizational, site requirements and related technical procedures
- KU2.** usage and handling procedure of related system components
- KU3.** tools involved in manufacturing of electrolyzer activities
- KU4.** functioning of different mechanical and electrical equipment, and required precautions to handle them

Qualification Pack

KU5. obtain relevant authorization

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation applicable to the role
- GS2.** read and correctly interpret health and safety and other technical guidelines
- GS3.** read from different sources- books, screens in machines and signage
- GS4.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS5.** express statements with required clarity
- GS6.** make timely decisions for efficient utilization of resources
- GS7.** follow organizational code of conduct

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Materials and components required for Manufacturing Electrolyzer</i>	20	30	-	-
PC1. explain and illustrate the basic materials and components required for manufacturing Electrolyzer and different types of Materials required for Electrolyzer manufacturing and their key specification through Pictures, videos, product data sheet etc.	2	6	-	-
PC2. explain the major components required for Alkaline type Electrolyzer. and discuss about each component.	2	-	-	-
PC3. explain the major components required for PEM type Electrolyzer and discuss about each component	3	-	-	-
PC4. explain the major components required for (SOEC) type Electrolyzer and discuss about each component.	3	-	-	-
PC5. prepare and Identify Bill of material for materials and components of Electrolyzer manufacturing set up.	6	-	-	-
PC6. explain the major components required for (AEM) type Electrolyzer and discuss about each component.	4	-	-	-
PC7. outline differences in PEM, AE, Electrolyzer Materials and components.	-	8	-	-
PC8. show how to identify Bill of material for materials and components of Electrolyzer manufacturing set up.	-	8	-	-
PC9. showcase the successful case studies of Electrolyzer technology for Green Hydrogen generation in India and overseas.	-	8	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4319
NOS Name	Identify Electrolyzer Components and Materials
Sector	Green Jobs
Sub-Sector	Green Hydrogen
Occupation	Hydrogen Plant Technician
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

SGJ/N4320: Perform Manufacturing Process of Electrolyzer

Description

This unit tells about Fundamental techniques and process for Manufacturing of Electrolyzer.

Scope

The scope covers the following :

- Electrolyzer cell configuration and its types
- Electrolyzer membrane selection and integration
- Explain electrode fabrication techniques of Electrolyzer.
- Explain step by step process of installation of Electrolyzer
- Explain assembling and Disassembling of Electrolyzer.

Elements and Performance Criteria

Fundamental techniques and process for manufacturing Electrolyzer

To be competent, the user/individual on the job must be able to:

- PC1.** identify and show different types of materials used in manufacturing electrolyzer.
- PC2.** discuss about the electrolyzer cell components and its types and show them the electrolyzer cell components and its types through pictures and videos.
- PC3.** discuss and show the process for membrane selection and Integration and how to select appropriate membrane for Electrolyzer manufacturing.
- PC4.** discuss about the stacking techniques for electrolyzer manufacturing.
- PC5.** discuss about Emerging Manufacturing Technologies for Electrolyzers.
- PC6.** discuss and show the scaling up and automation in electrolyzer manufacturing through pictures and videos.
- PC7.** discuss about the progress and Challenges in Electrolyzer Manufacturing industry.

Assembling and Disassembling of Electrolyzer

To be competent, the user/individual on the job must be able to:

- PC8.** explain and show how to read and interpret drawings of electrolyzer manufacturing facility
- PC9.** discuss and show how to perform Cleaning and inspection of components of Electrolyzer for wear and tear.
- PC10.** discuss step by step process for assembly/Installation of Electrolyzer- Bipolar plate, separator(membrane), Porous Transport layer, electrodes etc.
- PC11.** discuss and show the process of Membrane electrode assembly (MEA) handling and installation and process of membrane preparation through pictures and videos.
- PC12.** discuss and show how to implement the procedure for Electrolyzer stack assembly.
- PC13.** discuss the importance of Manifold and sealing in Electrolyzer Manufacturing
- PC14.** identify and showcase the key aspects and learnings of successful cases studies of Electrolyzer manufacturing plants in India and abroad.

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** precautions to be taken while handling different electrical and mechanical components
- KU2.** legislative, organizational, site requirements and technical procedures
- KU3.** • importance of wearing protective clothing and other safety gear while carrying out
• maintenance activities (PPEs)
- KU4.** mechanical and electrical equipment and their functioning
- KU5.** tools involved in assembling and disassembling of electrolyzer activities

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation work applicable to the role
- GS2.** follow organisation's code of conduct
- GS3.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- GS4.** follow organization's rule-based decision making process
- GS5.** make timely decisions for efficient utilization of resources

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fundamental techniques and process for manufacturing Electrolyzer</i>	20	30	-	-
PC1. identify and show different types of materials used in manufacturing electrolyzer.	2	6	-	-
PC2. discuss about the electrolyzer cell components and its types and show them the electrolyzer cell components and its types through pictures and videos.	3	8	-	-
PC3. discuss and show the process for membrane selection and Integration and how to select appropriate membrane for Electrolyzer manufacturing.	3	8	-	-
PC4. discuss about the stacking techniques for electrolyzer manufacturing.	4	-	-	-
PC5. discuss about Emerging Manufacturing Technologies for Electrolyzers.	4	-	-	-
PC6. discuss and show the scaling up and automation in electrolyzer manufacturing through pictures and videos.	2	8	-	-
PC7. discuss about the progress and Challenges in Electrolyzer Manufacturing industry.	2	-	-	-
<i>Assembling and Disassembling of Electrolyzer</i>	20	30	-	-
PC8. explain and show how to read and interpret drawings of electrolyzer manufacturing facility	3	6	-	-
PC9. discuss and show how to perform Cleaning and inspection of components of Electrolyzer for wear and tear.	3	6	-	-
PC10. discuss step by step process for assembly/Installation of Electrolyzer- Bipolar plate, separator(membrane), Porous Transport layer, electrodes etc.	3	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. discuss and show the process of Membrane electrode assembly (MEA) handling and installation and process of membrane preparation through pictures and videos.	3	6	-	-
PC12. discuss and show how to implement the procedure for Electrolyzer stack assembly.	3	6	-	-
PC13. discuss the importance of Manifold and sealing in Electrolyzer Manufacturing	3	-	-	-
PC14. identify and showcase the key aspects and learnings of successful cases studies of Electrolyzer manufacturing plants in India and abroad.	2	6	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4320
NOS Name	Perform Manufacturing Process of Electrolyzer
Sector	Green Jobs
Sub-Sector	Green Hydrogen
Occupation	Hydrogen Plant Technician
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

SGJ/N4321: Identify Tools and Tackles used in Electrolyzer Manufacturing Facility

Description

This unit tells about Identification of tools and Tackles used in Electrolyzer Manufacturing Facility.

Scope

The scope covers the following :

- Identify Common Tools Used in Electrolyzer Manufacturing facility
- tools required for performing electrical and mechanical work for assembling electrolyser.
- Factors to be consider when selecting tools and tackles for Electrolyzer manufacturing facility

Elements and Performance Criteria

Introduction to Tools and Tackles used in Electrolyzer Manufacturing Facility

To be competent, the user/individual on the job must be able to:

- PC1.** identify and illustrate the suitable tools and equipment required for electrolyzer manufacturing and classification of tools and tackles used in Electrolyzer Manufacturing facility.
- PC2.** prepare List of common tools and tackles used in electrolyzer manufacturing facility.
- PC3.** discuss and show the tools required for performing electrical and mechanical work for assembling electrolyzer and different types of tools and tackles used in electrolyzer manufacturing facility.
- PC4.** explain Factors to be consider when selecting tools and tackles for Electrolyzer manufacturing facility.
- PC5.** explain method for care of tools and tackles
- PC6.** discuss the function and attribute of each tool.
- PC7.** explain and show Storage, maintenance and repair tools used in electrolyzer manufacturing facility.
- PC8.** explain and Prepare list of frequent problems encountered with tools and provide solutions for troubleshooting.
- PC9.** discuss Checking and detection of damaged tools.
- PC10.** explain Disposal of damaged or worn-out tools.
- PC11.** discuss the Importance of using the right tools and tackles for the job.
- PC12.** discuss do's and don'ts activity with the use of tools in electrolyzer manufacturing industry.
- PC13.** explain Safety considerations for using power tools in the electrolyzer manufacturing facility.
- PC14.** discuss about best practices for care and maintenance of tools and tackles used in Electrolyzer Manufacturing facility.
- PC15.** identify and analyse the best practices for care and maintenance of tools and tackles used in Electrolyzer Manufacturing facility.in India and overseas.
- PC16.** show importance of calibration and traceability of measuring instruments

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** use of the tools and tackles
- KU2.** legislative, organizational, site requirements and procedures
- KU3.** how to obtain required authorization
- KU4.** precautions to be taken while handling different electrical and mechanical tools
- KU5.** usage and handling procedure of tools and tackles
- KU6.** cleaning procedures for various tools
- KU7.** reporting procedures

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** correctly read and interpret various color codes, as per standard electrical, mechanical and civil nomenclature
- GS2.** follow organizational rule-based decision making process
- GS3.** approach relevant authority when required
- GS4.** work constructively and collaboratively with others
- GS5.** express statements with required clarity
- GS6.** make timely decisions for efficient utilization of resources
- GS7.** follow organisational code of conduct

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Tools and Tackles used in Electrolyzer Manufacturing Facility</i>	25	25	-	-
PC1. identify and illustrate the suitable tools and equipment required for electrolyzer manufacturing and classification of tools and tackles used in Electrolyzer Manufacturing facility.	2	2	-	-
PC2. prepare List of common tools and tackles used in electrolyzer manufacturing facility.	-	4	-	-
PC3. discuss and show the tools required for performing electrical and mechanical work for assembling electrolyzer and different types of tools and tackles used in electrolyzer manufacturing facility.	1	4	-	-
PC4. explain Factors to be consider when selecting tools and tackles for Electrolyzer manufacturing facility.	2	-	-	-
PC5. explain method for care of tools and tackles	2	-	-	-
PC6. discuss the function and attribute of each tool.	2	-	-	-
PC7. explain and show Storage, maintenance and repair tools used in electrolyzer manufacturing facility.	1	4	-	-
PC8. explain and Prepare list of frequent problems encountered with tools and provide solutions for troubleshooting.	1	4	-	-
PC9. discuss Checking and detection of damaged tools.	2	-	-	-
PC10. explain Disposal of damaged or worn-out tools.	2	-	-	-
PC11. discuss the Importance of using the right tools and tackles for the job.	2	-	-	-
PC12. discuss do's and don'ts activity with the use of tools in electrolyzer manufacturing industry.	2	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. explain Safety considerations for using power tools in the electrolyzer manufacturing facility.	2	-	-	-
PC14. discuss about best practices for care and maintenance of tools and tackles used in Electrolyzer Manufacturing facility.	2	-	-	-
PC15. identify and analyse the best practices for care and maintenance of tools and tackles used in Electrolyzer Manufacturing facility.in India and overseas.	2	3	-	-
PC16. show importance of calibration and traceability of measuring instruments	-	4	-	-
NOS Total	25	25	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4321
NOS Name	Identify Tools and Tackles used in Electrolyzer Manufacturing Facility
Sector	Green Jobs
Sub-Sector	Green Hydrogen
Occupation	Hydrogen Plant Technician
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

SGJ/N4322: Perform Health and safety measures in Electrolyzer manufacturing facility.

Description

This unit tells about how to perform Perform Health and safety measures in Electrolyzer manufacturing facility.

Scope

The scope covers the following :

- Requirements for safe work area at Electrolyzer in hydrogen generation project site.
- Identify the hazards associated with Electrolyzer in a hydrogen generation system and their mitigation measures

Elements and Performance Criteria

Perform Health and safety measures in Electrolyzer manufacturing facility.

To be competent, the user/individual on the job must be able to:

- PC1.** explain the requirements for a safe work area at the Electrolyzer Manufacturing facility.
- PC2.** explain the importance of Occupational health & Safety standards and regulations for the safety of Electrolyzer Manufacturing plant facility.
- PC3.** describe potential causes of emergency such as gas leaks, fire, explosion, bomb threatening, natural calamities etc.
- PC4.** discuss importance of different detectors and safety tools used in Electrolyzer Manufacturing facility
- PC5.** review the Material Safety Data Sheet and labels of chemicals contained in cylinders in order to be aware of their hazards and precautionary measures
- PC6.** explain and demonstrate how to administer first aid and importance of first aid.
- PC7.** identify and demonstrate the usage of personal protective equipment for ensuring safety during manufacturing of Electrolyzer.
- PC8.** identify the hazards associated with Electrolyzer Manufacturing and show how to comply with all applicable statutory requirements along with safety regulations in terms of fire protection.
- PC9.** identify work safety procedures and instructions for working at Electrolyzer Manufacturing facility.
- PC10.** discuss Mock testing of firefighting system and demonstrate the use of fire extinguishers, fire detection and alarm system.
- PC11.** discuss all applicable statutory requirements along with safety regulations in terms of fire protection and demonstrate how to follow necessary and adequate safety measures including personal protective equipment and precautions to avoid any accident at electrolyzer manufacturing site.
- PC12.** incorporate and demonstrate good housekeeping and infection control & prevention practices.

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- PC13.** identify and showcase best practices for implementing health and safety measures in manufacturing of electrolyzer in India and overseas.
- PC14.** demonstrate the use of fire extinguishers, fire detection and alarm system.
- PC15.** demonstrate how to follow necessary and adequate safety measures including personal protective equipment and precautions to avoid any accident at electrolyzer manufacturing site.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** complete documentation work applicable to the role
- KU2.** work constructively and collaboratively with others
- KU3.** follow organization's rule-based decision making process
- KU4.** make timely decisions for efficient utilization of resources
- KU5.** read various color codes, as per standard electrical, mechanical and civil nomenclature
- KU6.** read and correctly interpret health and safety instructions and signage

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** obtain authorization from specified officials
- GS2.** legislative, organization, site requirements and technical procedures
- GS3.** usage and handling procedure related to safety of electrolyzer manufacturing facility
- GS4.** importance of wearing protective clothing and other safety gear
- GS5.** read statutory documents relevant to health and safety

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform Health and safety measures in Electrolyzer manufacturing facility.</i>	25	25	-	-
PC1. explain the requirements for a safe work area at the Electrolyzer Manufacturing facility.	2	-	-	-
PC2. explain the importance of Occupational health & Safety standards and regulations for the safety of Electrolyzer Manufacturing plant facility.	2	-	-	-
PC3. describe potential causes of emergency such as gas leaks, fire, explosion, bomb threatening, natural calamities etc.	2	-	-	-
PC4. discuss importance of different detectors and safety tools used in Electrolyzer Manufacturing facility	2	-	-	-
PC5. review the Material Safety Data Sheet and labels of chemicals contained in cylinders in order to be aware of their hazards and precautionary measures	3	-	-	-
PC6. explain and demonstrate how to administer first aid and importance of first aid.	1	5	-	-
PC7. identify and demonstrate the usage of personal protective equipment for ensuring safety during manufacturing of Electrolyzer.	1	5	-	-
PC8. identify the hazards associated with Electrolyzer Manufacturing and show how to comply with all applicable statutory requirements along with safety regulations in terms of fire protection.	2	5	-	-
PC9. identify work safety procedures and instructions for working at Electrolyzer Manufacturing facility.	2	-	-	-
PC10. discuss Mock testing of firefighting system and demonstrate the use of fire extinguishers, fire detection and alarm system.	2	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. discuss all applicable statutory requirements along with safety regulations in terms of fire protection and demonstrate how to follow necessary and adequate safety measures including personal protective equipment and precautions to avoid any accident at electrolyzer manufacturing site.	2	3	-	-
PC12. incorporate and demonstrate good housekeeping and infection control & prevention practices.	2	2	-	-
PC13. identify and showcase best practices for implementing health and safety measures in manufacturing of electrolyzer in India and overseas.	2	2	-	-
PC14. demonstrate the use of fire extinguishers, fire detection and alarm system.	-	-	-	-
PC15. demonstrate how to follow necessary and adequate safety measures including personal protective equipment and precautions to avoid any accident at electrolyzer manufacturing site.	-	-	-	-
NOS Total	25	25	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4322
NOS Name	Perform Health and safety measures in Electrolyzer manufacturing facility.
Sector	Green Jobs
Sub-Sector	Green Hydrogen
Occupation	Hydrogen Plant Technician
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council/Awarding Body. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC/AB will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC/AB.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the assessment, every trainee should score the Recommended Pass % aggregate for the Qualification.

Qualification Pack

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N4318.Introduction about the Electrolyzer and its types	40	60	0	0	100	18
SGJ/N4319.Identify Electrolyzer Components and Materials	20	30	0	0	50	16
SGJ/N4320.Perform Manufacturing Process of Electrolyzer	40	60	0	0	100	18
SGJ/N4321.Identify Tools and Tackles used in Electrolyzer Manufacturing Facility	25	25	0	0	50	16
SGJ/N4322.Perform Health and safety measures in Electrolyzer manufacturing facility.	25	25	0	0	50	16
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	16
Total	170	230	-	-	400	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.